MILITARY SPECIFICATION

COLISTIN SULFATE, HYDROCORTISONE ACETATE, NEOMYCIN SULFATE, AND THONZONIUM BROMIDE SUSPENSION, OTIC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers Colistin Sulfate, Hydrocortisone Acetate, Neomycin Sulfate and Thonzonium Bromide Suspension, Otic (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-C-186 - Containers, Packaging and Packing for Drugs, Chemicals and Pharmaceuticals.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Headquarters, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, DPSC-ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6505

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

U. S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE,
FOOD AND DRUG ADMINISTRATION (FDA)

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.

(Application for copies should be addressed to the Food and Drug Administration, Washington, DC 20204.)

U. S. PHARMACOPEIAL CONVENTION, INC.

The United States Pharmacopeia (USP).

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

The National Formulary (NF).

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

3. REQUIREMENTS

3.1 Material. Shall be a sterile, non-viscous, aqueous suspension containing in each ml, 3.0 mg colistin base (present as the sulfate), 3.3 mg neomycin base (present as the sulfate), 10.0 mg hydrocortisone acetate, 0.5 mg thonzonium bromide, and a suitable preservative.

3.2 Assay.

3.2.1 Colistin and neomycin. The otic suspension shall contain within applicable FDA limits, the required amounts of colistin and neomycin, when assayed by the methods included in the Food and Drug Administration regulations.

3.2.2 Hydrocortisone acetate and thonzonium bromide. The otic suspension shall assay to contain not less than 90.0 percent and not more than 110.0 percent of the required amounts of hydrocortisone acetate and thonzonium bromide, when assayed as specified in 4.7.1 and 4.7.2, respectively.

3.3 Sterility. The otic suspension shall meet the test requirements of the FDA for sterility.

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3.4 Ingredients.

3.4.1 Colistin sulfate and neomycin sulfate. The colistin sulfate and neomycin sulfate used in the manufacture of the suspension shall comply with the tests, standards, and requirements of the USP, including any supplements or revisions thereto and the applicable regulations for the Certification of Antibiotics and Antibiotic-Containing Drugs, as promulgated by the FDA.

3.4.2 Hydrocortisone acetate. The hydrocortisone acetate used in the manufacture of the suspension shall comply with the tests, standards, and requirements of the USP, including any supplements or revisions thereto.

3.4.3 Thonzonium bromide. The thonzonium bromide used in the manufacture of the suspension shall meet the following requirements:

3.4.3.1 pH. Shall be not less than 5.0 and not more than 6.5 at 25° C, for a 1.0 percent solution in distilled water, when determined potentiometrically, using a glass electrode.

3.4.3.2 Identity. When treated with an excess of saturated picric acid solution, a 1.0 percent solution in purified water shall yield a yellow precipitate.

3.4.3.3 Melting point. Shall be not less than 93° C and not more than 97° C, when determined in accordance with the USP procedure for class I.

3.4.3.4 Moisture. Shall be not more than 0.5 percent when determined as follows:

Weigh, accurately, about 2 gram of sample. Place in a desiccator over alumina or anhydrone at room temperature (25° C) for 24 hours. Remove and weigh.

3.4.3.5 Assay. Shall assay to contain not less than 98.0 percent and not more than 102.0 percent (on a dry basis) of thonzonium bromide when determine as specified in 4.7.3.

3.5 Other ingredients. All other ingredients entering into the preparation shall be of USP or NF quality, or if not included in either of these compendia, they shall be suitable for use in the preparation.

3.6 Certification. The finished otic suspension shall be in accordance with the General Regulations for the Certification of Antibiotics and Antibiotic-containing Drugs, as promulgated by the FDA. Each lot shall be certified by the FDA.

3.7 Expiration dating period. The expiration dating period shall be 18 months.

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3.7.1 Delivery. Not more than 2 months of the expiration dating period shall have elapsed at the time of delivery to the Government.

3.8 Workmanship. The material and its containers shall be free from defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and, (i) as to any expiration dated item, for the expiration dating period specified by the contractor for such item, or such longer period as may be required by regulation of any federal agency; or (ii) as to non-dated items, for not less than 3 years after delivery of the item to the Government.

4.1.2 No company supplying any ingredient(s) to the contractor will be considered an acceptable facility for the performance of any inspection requirements specified herein.

4.2 Lot. For purposes of this specification, a lot, batch, or control is that single, uniform, and homogeneous quantity of suspension produced from one formulation, subjected to the same compounding and manufacturing operation, and filled into final containers.

4.3 Sampling. Sampling shall be conducted in accordance with MIL-STD-105.

TABLE I. Sampling.

For visual examination	Inspection level	AQL (percent defective)	Unit of product
Major	S-2	1.0	Filled immediate container
Minor	II	2.5	Unit package

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4.3.1 Raw material and end item testing. Noncompliance with any of the following is cause for rejection.

TABEE II. Criteria for raw material and end item testing.

Characteristic	Requirement paragraph	Inspection level 1/	Minimum sample
Assay:			
Colistin and Neomycin	3.2.1	(FDA regulations)	
Hydrocortisone Acetate and Thonzonium Bromide	3.2.2	S-2	5 (composite)
Ingredients: 2/			
Thonzonium Bromide	3.4.3	S-2	-

1/ Where possible, the sample shall be used for two or more tests.
2/ Sufficient quantities of each of the raw materials shall be selected from each lot to perform tests indicated in applicable test methods. Failure of any sample to comply with test requirements shall be cause for rejection of the lot represented.

4.4 Examination. The finished suspension shall be examined to determine compliance with all requirements of this specification. Nonconformance will be permitted to the extent indicated in 4.3 and 4.3.1.

4.4.1 Classification of defects. Examination will be conducted in accordance with the following classification of defects:

TABLE III. Classification of defects.

Categories	Defects 1/
Major	
101	Suspension not free from foreign matter.
102	Suspension not homogeneous.
Minor	
201	Exterior of immediate container is not free from contents or foreign matter.
202	Label not free from defacing marks and tears.

1/ Inspection is not restricted to the classified possible defects listed above.

4.5 Packaging inspection. The inspection of the intermediate package, packing and marking for shipment and storage shall be in accordance with quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

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4.6 Labeling inspection. Sampling shall be in accordance with MIL-STD-105 and the unit of product shall be the unit of issue. The inspection lot shall be the number of units of issue of each batch, lot, control number, etc., offered for inspection at one time. The inspection level for labeling shall be S-2 and the acceptance number shall be zero.

4.7 Assay.

4.7.1 Hydrocortisone acetate.

Reagents.

(a) Dilute H_2SO_4: Add 620 ml of concentrated H_2SO_4 to 380 ml of purified water.

(b) Reagent Blank: Dilute 200 ml of dilute H_2SO_4 (reagent (a)) with 100 ml of absolute alcohol.

(c) Phenylhydrazine Reagent: Dissolve 43.33 mg of phenylhydrazine hydrochloride in 100 ml of reagent blank (reagent (b)).

Preparation of standard solution. Weigh, accurately, 100 mg of hydrocortisone acetate USP reference standard, transfer into a 500 ml volumetric flask, and bring to volume with chloroform ($CHCl_3$). Pipet 5 ml of the above solution into a 100 ml volumetric flask, and dilute to volume with $CHCl_3$.

Preparation of sample solution. Pipet 5 ml of a well mixed (shake vigorously) sample into a 125 ml separatory funnel. Extract with three 20 ml volumes of $CHCl_3$, transferring all $CHCl_3$ layers through a pledget of cotton, previously saturated with $CHCl_3$, into a 100 ml volumetric flask. Dilute the $CHCl_3$ extracts to volume with $CHCl_3$. Pipet 10 ml of this solution into a second 100 ml volumetric flask and dilute to volume with $CHCl_3$. Pipet 20 ml of this solution into a third 100 ml volumetric flask and dilute to volume with $CHCl_3$.

Procedure. Pipet 50 ml of the standard solution and 50 ml of the sample solution into respective 125 ml separatory funnels, add 2 ml of 0.1N NaOH to each funnel, shake and let solutions separate. Filter both $CHCl_3$ layers through glass wool into separate beakers. Pipet two 20 ml volumes of the $CHCl_3$ filtrate from each beaker into four 125 ml separatory funnels. Add 25 ml of reagent blank (reagent (b)) to one set of separatory funnels and 25 ml of phenylhydrazine reagent (reagent (c)) to the other set of separatory funnels. Shake well, allow layers to separate, and discard the $CHCl_3$ layers. Transfer the aqueous layers into centrifuge tubes and centrifuge for 2 minutes. Pipet 10 ml of each sample into respective ground glass test tubes, and heat in a water bath at 60° C $\pm$ 2° C for 30 minutes. Remove from the water bath, cool to room temperature, and determine the absorbancies of the four solutions in a Beckman Spectrophotometer, Model DU, or equivalent instrument giving comparable results, at 410 nm, using 1-cm silica cells, against purified water.

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Calculation:

$$\frac{A - B}{C - D} \times \frac{20 \times 10 \times 5}{5 \times 20} = \text{mg hydrocortisone acetate per ml}$$

Where: A = Sample absorbance.
B = Sample blank absorbance.
C = Standard absorbance.
D = Standard blank absorbance.

4.7.2.1 Thonzonium bromide.

Reagents.

Bromthymol blue solution. Weigh 150 mg of bromthymol blue and 150 mg of sodium carbonate (Na_2CO_3), anhydrous, transfer into a 100 ml graduate and dilute to volume with purfied water.

Sodium carbonate buffer. Weigh 20.0 Gram of Na_2CO_3, anhydrous, transfer into a 100 ml graduate and dilute to volume with purified water.

Boric acid solution. Weigh 5.0 Gram boric acid and transfer to a 250 ml volumetric flask. Add 200 ml absolute alcohol and swirl to complete solution. Dilute to volume with absolute alcohol.

Thonzonium bromide standard. Weigh, accurately, 50.0 mg of the thonzonium bromide, transfer into a 500 ml volumetric flask and dilute to volume with purified water.

Procedure. Pipet 5 ml of a well mixed sample into a 25 ml volumetric flask and dilute to volume with purified water. Pipet 15 ml of this sample dilution into a 125 ml separatory funnel and 15 ml of the thonzonium bromide standard into a second 125 ml separatory funnel. Add 20 ml of chloroform ($CHCl_3$), 2.0 ml of bromthymol blue solution and 2.0 ml of Na_2CO_3 buffer solution to each funnel and shake vigorously for 2 minutes. Allow layers to separate and transfer the $CHCl_3$ layers to respective 125 ml glass-stoppered Erlenmeyer flasks, each containing about 0.4 Gram of glass wool. Repeat the extractions with two additional 20 ml volumes of $CHCl_3$, adding each to the respective 125 ml Erlenmeyer flasks. Shake the $CHCl_3$ extractions for about 1 minute and let settle for 5 minutes. Decant the $CHCl_3$ extraction through glass wool into 100 ml volumetric flasks, each containing 25 ml of boric acid solution. Rinse the 125 ml Erlenmeyer flasks free of yellow color with small portions of $CHCl_3$, adding these rinsings to the 100 ml volumetric flasks to make 100 ml and mix thoroughly. As a blank measure 25 ml of boric acid solution into a 100 ml volumetric flask, dilute to volume with $CHCl_3$ and mix thoroughly. Determine the absorbancies of the solutions in a Beckman Spectrophotometer, Model B, or equivalent instrument giving comparable results, at 420 nm, using 1-cm corex or silica cells. Set the instrument at zero absorbance with $CHCl_3$.

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Calculation:

$$\frac{A - C}{B - C} \times \frac{50}{500} \times \frac{15}{100} \times \frac{25}{5} \times \frac{100}{15} = \text{mg thonzonium bromide per ml.}$$

Where: A = Sample absorbance.
B = Standard absorbance.
C = Sample blank absorbance.

4.7.3 Thonzonium bromide (ingredient).

Weigh, accurately, about 0.75 Gram of sample into a 250 ml Erlenmeyer flask. Add about 80 ml of glacial acetic acid and swirl until solution is clear. Then add mercuric acetate TS until the precipitate first formed dissolves. Add an excess of 10 ml mercuric acetate TS, two drops of 1 percent crystal violet solution and titrate with 0.1N perchloric acid to a blue-green endpoint. Each ml of 0.1N perchloric acid equals 0.02959 Gram thonzonium bromide.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with all applicable requirements of PPP-C-186, and as specified herein.

5.2 Packaging. Packaging shall be level A or C, as specified (see 6.1).

5.2.1 Unit of issue. One package containing one bottle and one sterile dropper, as specified, constitutes one unit of issue.

5.2.2 Level A.

5.2.2.1 Immediate container. Immediate container shall be a commercially available immediate container considered acceptable by the Food and Drug Administration (FDA) for the product contained therein and shall be of such compatibility that neither the contents nor the container are altered in any way by each other. In addition, the immediate container shall be furnished with a separate sterile dropper.

5.2.2.2 Unit package. Each unit shall be packaged as specified in 5.2.5 of PPP-C-186.

5.2.2.3 Intermediate package. Twelve (12) unit packages shall be packaged in an intermediate package as specified in 5.2.7.1 of PPP-C-186.

5.2.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

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5.3 Packing. Packing shall be level A, B or C, as specified (see 6.1).

5.3.1 Level A. Twenty-four intermediate packages (288 units) shall be packed in an exterior container as specified in 5.4.4 of PPP-C-186.

5.3.2 Level B. Twenty-four intermediate packages (288 units) shall be packed in an exterior container as specified in 5.4.3.1.1 of PPP-C-186.

5.3.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.4 Marking.

5.4.1 Labeling. Labeling shall be in accordance with the requirements of the Federal Food, Drug and Cosmetic Act, and shall include the information required below.

5.4.1.1 Immediate container. Each immediate container label shall bear the following information:

(a) label information in accordance with commercial practice.
(b) the expiration date.

5.4.1.2 Unit package. Each unit package label shall bear the following information:

(a) label information in accordance with commercial practice.
(b) the national stock number.
(c) extended potency lines.

NOTES: (1) The national stock number shall not be preceded by any prefix.
(2) Strip labeling may be used.
(3) Label adhesion test shall not apply.

5.4.2 Intermediate package. Each intermediate package shall be marked in accordance with 5.5.3 of PPP-C-186. Type II shelf life marking as specified in MIL-STD-129 shall include the legend: "STORE AT CONTROLLED ROOM TEMPERATURE 15° - 30° C (59° - 86° F)".

5.4.3 Exterior container. Exterior containers shall be marked as specified in 5.5.4 of PPP-C-186. Type II shelf life marking as specified in MIL-STD-129 shall be shown. Marking shall include the legend: "STORE AT CONTROLLED ROOM TEMPERATURE 15° - 30° C (59° - 86° F)".

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6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) the National Stock Number.
(c) Selection of applicable levels of packaging and packing (see 5.2 and 5.3).

6.2 This specification covers the following item(s):

National Stock Number	Item Identification
6505-00-890-1907	COLISTIN SULFATE, HYDROCORTISONE ACETATE, NEOMYCIN SULFATE, AND THONZONIUM BROMIDE SUSPENSION, OTIC, 5 ml

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activity:
DoD - MB

Preparing activity:
DLA - DM

Project No. 6505-1735

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OMB Approval
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DOCUMENT IDENTIFIER AND TITLE MIL-C-37590 COLISTIN SULFATE, HYDROCORTISONE ACETATE, NEOMYCIN SULFATE, AND THONZONIUM BROMIDE SUSPENSION, OTIC

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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